Low-Rank Tensor PDE Dissertation Showcase

ReproMath example

1 Project Overview

This miniature dissertation project demonstrates how ReproMath can keep a mathematical research workflow checkable. A low-rank tensor PDE project often connects tensor notation, HOSVD-style approximations, numerical diagnostics, figures, and final LaTeX writing. The purpose of this example is to keep those artifacts small, reproducible, and mapped in `repromath.toml`.

2 Tensor Notation

For a third-order tensor, a mode unfolding rewrites multiway data as a matrix view. This is not a change in the underlying data; it is a bookkeeping device that lets matrix tools such as the singular value decomposition act on one mode at a time.

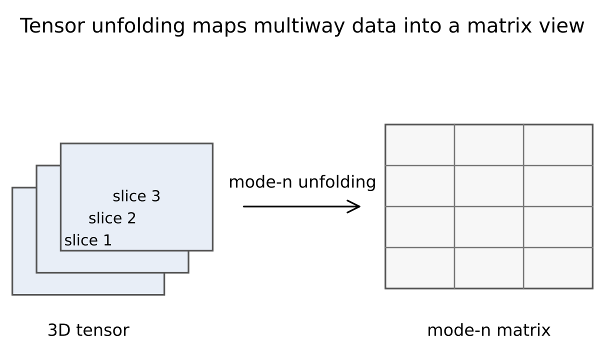

Figure 1: A conceptual mode unfolding used to connect tensor notation with matrix computations.

In a truncated HOSVD study, the notebook records the factor matrices, the core tensor, and the diagnostics needed to decide whether a chosen multilinear rank is numerically reasonable.

3 Numerical Diagnostics

The first question in a low-rank approximation experiment is whether the singular values decay quickly enough to justify truncation. Figure 2 shows a toy comparison between fast and slow decay.

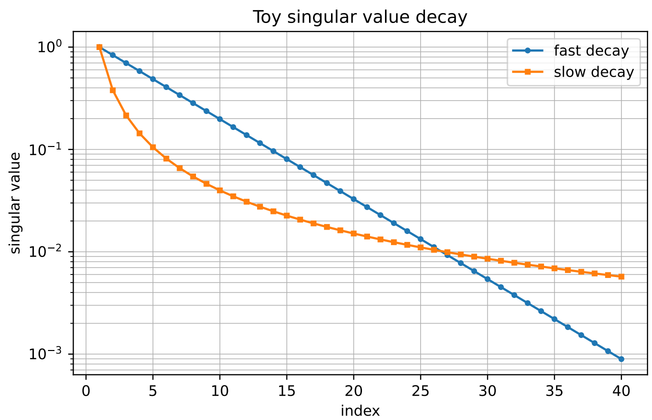

Figure 2: Toy singular value decay curves used as a diagnostic for low-rank approximation.

The second question is whether the compressed representation saves enough storage to justify the approximation. Figure 3 compares full tensor storage with a simple equal-rank Tucker storage count.

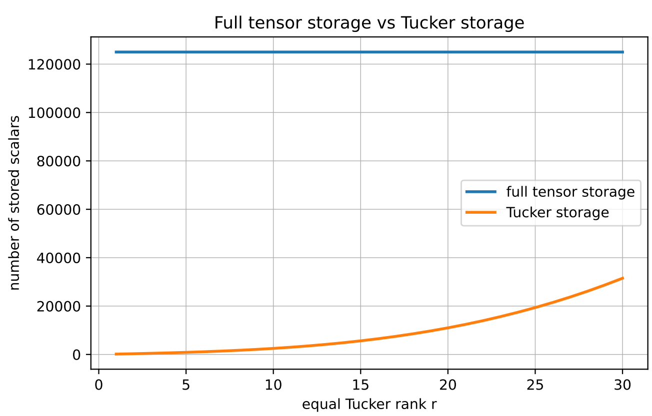

Figure 3: A lightweight storage comparison for a cubic tensor and equal Tucker rank.

The corresponding notebook keeps these diagnostics close to the code that generates them, so that dissertation figures and computational checks can be reviewed together.